

Abhijit Shankar Mudigonda

CONTACT INFORMATION	2624 Montebelluna Pl Leander, TX 78641 (971)-282-7028	abhijit.mudigonda@gmail.com abhijit-mudigonda  https://abhijit-mudigonda.github.io/math
RESEARCH INTERESTS	Number Theory, Complex Geometry, Theoretical Computer Science	
EDUCATION	University of Chicago Ph.D. Candidate in Mathematics and Computer Science, September 2021 - Present M.S. in Computer Science, June 2024 Massachusetts Institute of Technology B.S. in Mathematics, September 2015 - June 2019	
PUBLICATIONS	A. S. Mudigonda. <i>Computing Hilbert modular forms with nonparititious weight</i> . Proceedings of the Algorithmic Number Theory Symposium XVII, to appear. B. Matschke and A. S. Mudigonda. <i>Quadratic Fields Admitting Elliptic Curves with Rational j-Invariant and Good Reduction Everywhere</i> . Journal of Number Theory. Volume 247, June 2023, Pages 162-210. A. S. Mudigonda and R. R. Williams. <i>Time-Space Lower Bounds for Simulating Proof Systems with Quantum and Randomized Verifiers</i> . Proceedings of the 2021 Conference on Innovations in Theoretical Computer Science.	
PREPRINTS	E. Assaf, A. S. Mudigonda. <i>Computing Hilbert modular forms with nontrivial nebensystem</i> . In preparation. A. S. Mudigonda. <i>Galois representations of abelian varieties with nontrivial geometric endomorphisms</i> . In preparation. E. Assaf, A. Babei, B. Breen, S. Chari, E. Costa, J. Duque-Rosero, A. Horawa, J. Kieffer, A. Kulkarni, A. S. Mudigonda, M. Musty, S. Schiavone, S. Sethi, S. Tripp, J. Voight. <i>Computing with Fourier expansions of Hilbert modular forms</i> . In preparation.	
PATENTS	D. Gutfreund, Q. Fan, A. Mudigonda. <i>Restructuring Deep Neural Networks to Reduce the Number of Parameters</i> , USPA #20200160144. Published May 21, 2020.	
INVITED TALKS	January 2027 – JMM Special Session, Explicit Methods in Number Theory January 2027 – JMM Special Session, Mathematics Informed by Computing December 2026 – UW-Madison Number Theory Seminar July 2026 – Algorithmic Number Theory Symposium XVII May 2026 – UChicago Number Theory Seminar April 2026 – UC Santa Barbara Number Theory Seminar January 2021 – Innovations in Theoretical Computer Science January 2021 – UT Austin Quantum Information Seminar	

WORK EXPERIENCE	08/2020 - 09/2021	<i>Software Engineer, Facebook Reality Labs</i> Worked as an OS engineer on the XROS platform team, developing features for the Oculus Quest Pro.
	06/2018 - 08/2018	<i>Undergraduate researcher, MIT CSAIL</i> Studied uniform versions of algebraic circuit classes in complexity theory.
	06/2017 - 08/2017	<i>Software Engineering Intern, Facebook</i> Worked on ZippyDB, a persistent distributed key-value service built on top of RocksDB.
	01/2017 - 04/2017	<i>Research Intern, IBM Research.</i> Developed neural network architectures inspired by expander graphs.
	06/2016 - 08/2016	<i>Research Intern, Knight Cancer Institute.</i> Developed deep learning models to predict DNA sequence accuracy.
TEACHING EXPERIENCE	09/2025 - Present	<i>Teaching Assistant</i> Was a TA for Honors Introduction to Discrete Mathematics, Introduction to Algorithms, and Honors Introduction to Complexity Theory.
	09/2022 - 06/2023	<i>UChicago Teaching Fellow</i> Underwent a year-long training program preparing graduate students to be independent lecturers. Was a TA for Algebra I, Introduction to Riemannian Geometry, and Introduction to Algebraic Number Theory.
	06/2020 - 08/2020	<i>Summer STEM Institute Research Mentor</i> Mentored two high school students in theoretical and applied quantum computing research projects.
	01/2019	<i>MIT Global Teaching Labs Teacher</i> Taught physics and biology to high school students at the Liceo Scientifico Michelangelo Grigoletti in Pordenone, Italy.
	06/2015 - 06/2018	<i>USA Biology Olympiad Teaching Assistant</i> Wrote practical and theoretical examinations to select the 2015, 2016, 2017, and 2018 U.S. representative teams. Gave lectures on topics ranging from developmental physiology to biostatistics.
	11/2015 - 11/2016	<i>MIT Educational Studies Program Teacher</i> Taught courses in spectral graph theory and a cappella arranging
REFEREING	Algebra & Number Theory	
HONORS AND AWARDS	2021 NSF Graduate Research Fellow	
	2021 UChicago Crerar Fellow	
	2015 North American Computational Linguistics Olympiad - U.S. Alternate (11 th)	
	2014 International Biology Olympiad - Silver Medal (25 th)	
	2014 United States National Chemistry Olympiad - High Honors (Top 50)	